

Liyin Zhang

Ph.D. Student, Educational Technology

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Education & Training

Ph.D. Student	<i>Educational Technology, Zhejiang University, Hangzhou, 2022.9–2027.6 (expected)</i> Supervisor: Ouyang Fan, Chen Wenzhi (College of Computer Science and Technology), Ji Shouling (College of Computer Science and Technology), Liu Zhengwei
Visiting Student	<i>Educational Neuroscience, Nanyang Technological University, Singapore, 2025.8–2026.8</i> Supervisors: Teo Wei-Peng, Dragan Gasevic
B.A.	<i>Education, Zhejiang University, Hangzhou, 2018.9–2022.6</i> Bachelor of Education, National Scholarship, Outstanding Graduates of Zhejiang Province

Research Areas

- **Computer-Supported Collaborative Learning:** Collaborative knowledge construction; Collaborative learning analytics; Collaborative scaffolding; Group cognition and regulation.
- **AI and Multimodal Learning Analytics:** Multimodal learning analytics; AI-driven learning analytics; Computational modeling; Generative AI and intelligent learning support.
- **Educational Neuroscience:** fNIRS; Neural mechanisms of learning; Dynamic functional connectivity; Interpersonal brain synchronization.

Publications

C=Conference, J=Journal, S=Submitted/In Submission

* indicates cofirst author. † indicates the corresponding author.

- [J.1] Zhang, L., Wang, X., & Chen, W.[†], & Ji, S.[†], & Liu, Z.[†], & Ouyang, F.[†] (2026). Using multimodal learning analytics to understand the combined effects of dynamic self-regulated scaffoldings and learning from examples on computer programming: an empirical study. *International Journal of STEM Education*. DOI: 10.1186/s40594-025-00591-x. (SSCI, IF=10.7)
- [J.2] Zhang, L., Zhang, N., He, J., Guo, M., & Ouyang, F.[†] (2026). The influences of three types of peer-questioning scaffoldings on online peer learning in a virtual museum environment. *Interactive Learning Environments*. DOI: 10.1080/10494820.2025.2494150. (SSCI, IF=6.7)
- [J.3] Ouyang, F.[†], & Zhang, L.[†] (2024). AI-driven learning analytics applications and tools in computer-supported collaborative learning: A systematic review. *Educational Research Review*, 44, 100616. DOI: 10.1016/j.edurev.2024.100616. (SSCI, IF=10.3)
- [J.4] Zhang, L., Wu, M., & Ouyang, F.[†] (2024). The design and implementation of a teaching and learning analytics tool in a face-to-face, small-sized course in China's higher education. *Education and Information Technologies*. DOI: 10.1007/s10639-023-11940-0. (SSCI, IF=7.2)
- [J.5] Xia, L., Zhang, L., Shen, K., & Dong, Y.[†] (2026). The impact of generative artificial intelligence (GenAI) on university students' learning agency: A mixed-methods study. *Teaching in Higher Education*. DOI: 10.1080/13562517.2026.2649818. (SSCI, IF=3.9)
- [J.6] Ouyang, F.[†], Zhang, L., Wu, M., & Jiao, P. (2024). Empowering collaborative knowledge construction through the implementation of a collaborative argument map tool. *The Internet and Higher Education*, 62, 100946. DOI: 10.1016/j.iheeduc.2024.100946. (SSCI, IF=10.6)
- [J.7] Ouyang, F.[†], Wu, M., Zhang, L., Xu, W., Zheng, L., & Cukurova, M.[†] (2023). Making strides towards AI-supported regulation of learning in collaborative knowledge construction. *Computers in Human Behavior*, 142, 107650. DOI: 10.1016/j.chb.2023.107650. (SSCI, IF=12.2)
- [J.8] Ouyang, F., Wu, M., Zheng, L., Zhang, L., & Jiao, P.[†] (2023). Integration of artificial intelligence performance prediction and learning analytics to improve student learning in online engineering course. *International Journal of Educational Technology in Higher Education*, 20(4), 1–23. DOI: 10.1186/s41239-022-00372-4. (SSCI, IF=38.7)
- [J.9] Zhang, L., Lu, J., & Liu, H.[†] (2022). How can the cognitive skills of high school students be assessed? Developing a college-readiness performance assessment system (in Chinese). *Shanghai Education*, (11), 38–40.
- [J.10] Zhang, L., & Liu, H.[†] (2021). Encountering authentic assessment through rich tasks in Queensland (in Chinese). *Shanghai Education*, (2), 62–65.

- [J.11] Shen, Y., Li, Y., & **Zhang, L.** (2021). [Russia: Integrating professional orientations into high school curriculum and instruction \(in Chinese\)](#). *Shanghai Education*, (24), 12–15.
- [S.1] **Zhang, L.**, Xia, L., & Teo, W.-P.[†] (2026). Beyond gaming performance: Exploring the dynamic neural mechanisms underlying learning outcomes in game-based learning. *NeuroImage*. Awaiting Final Decision.
- [S.2] Xia, L.^{*}, Ho, M.-Y. M.^{*}, **Zhang, L.^{*}**, Chen, W., & Teo, W.-P. (2026). The impact of competition and awareness tools on emotion, cognitive load, and functional connectivity in digital game-based learning. *Journal of Computer Assisted Learning*. Major Revision.
- [S.3] **Zhang, L.**, Xia, L., Gasevic, D.[†], & Teo, W.-P.[†] (2026). Unpacking collaborative engagement through embodied data in multimodal learning analytics: A systematic review. *Educational Research Review*. Under Review.
- [S.4] Ouyang, F.^{*}, **Zhang, L.^{*}**, Yang, Y., Liu, X.[†], Ji, S., Chen, W., & Liu, Z. (2026). Automatic content analysis of collaborative knowledge construction using RoBERTa-CKC: A pretrained LLM approach. *IEEE Transactions on Learning Technologies*. Under Review.
- [S.5] **Zhang, L.**, Ji, S., Chen, W., Liu, Z., & Ouyang, F.[†] (2026). When generative artificial intelligence (GenAI) coordinates with pedagogical strategies: Unpacking programming learning dynamics through behaviors, learning performance and perceptions. *Educational Technology Research and Development*. Under Review.
- [S.6] Chen, S., Yan, M., Wu, J., & **Zhang, L.[†]** (2026). Empowering programming learning through a multi-agent system integrating cognitive scaffolding and affective feedback. *Asia Pacific Journal of Education*. Under Review.
- [S.7] **Zhang, L.**, Xia, L., Gasevic, D.[†], & Teo, W.-P.[†]. (2026). Educational neuroscience evidence for interpersonal and human- computer interactions: A systematic review of fNIRS research in higher education. *Educational Psychology Review*. In submission.
- [S.8] Xia, L., Li, X., Dong, Y., **Zhang, L.[†]**. (2026). When technology affordance does not guarantee learning: The role of learning agency across GenAI-supported and conventional search engine-supported learning. *Education and Information Technologies*. In submission.
- [C.1] **Zhang, L.**, Zhang, N., He, J., Guo, M., Ouyang, F.[†] (2025, May). [Enhancing online peer learning in a virtual museum: Investigating the impact of three peer-questioning scaffoldings](#). In *International Conference on Learning and Teaching 2025 (ICLT 2025)*. HongKong, China. Oral presentation.
- [C.2] **Zhang, L.**, Ouyang, F.[†] (2024, September). [Design and application of AI-driven learning analytics tools in computer-supported collaborative learning: A systematic review](#). In *the 19th Annual Conference of the Information Technology Education Professional Committee*. Chinese Association for Educational Technology, Shanghai, China. Online presentation.
- [C.3] **Zhang, L.**, Wu, M., Ouyang, F.[†] (2023, May). [The effects of a collaborative argument mapping tool on collaborative knowledge construction](#). In *the 18th Annual Conference of the Information Technology Education Professional Committee*. Chinese Association for Educational Technology, Shanghai, China. Online presentation.
- [C.4] **Zhang, L.**, Wu, M., Ouyang, F.[†] (2022, May). [The effects of a learning analytics visualization feedback tool on teaching and learning in higher education classrooms](#). In *the 26th Global Chinese Conference on Computers in Education (GCCCE 2026)*. Online presentation.
- [C.5] Xia, L., Ho, M.-Y. M., **Zhang, L.^{*}**, & Teo, W.-P.[†] (2025, November). [The impact of competition and awareness tools on emotion, cognitive load, and functional connectivity in digital game-based learning](#). In *14th Australasia Cognitive Neuroscience Society Conference*. Melbourne, Australia. Poster.

Professional Experience

2022-present Graduate Student Researcher, College of Education, Zhejiang University

- Projects:
- 2025.06 – 2026.06. Zhejiang University Interdisciplinary Program for Doctoral Students (Grant No. 188310-5421B3). Research on Key Technologies and Practical Applications of Generative Artificial Intelligence in Computer-Supported Collaborative Writing. *Host*
 - 2021.06 – 2022.06. Zhejiang University Undergraduate Student Research Training Program (SRTP) (Grant No. X20200039). Student–Student Interaction in Discussion Forums on Online Education Platforms. *Host*
 - 2022.01 – 2025.12. National Natural Science Foundation of China (NSFC), General Program (Grant No. 62177041). Mechanisms Underlying the Development of Group Cognition in Online Collaborative Learning: Computational Modeling, Analytic Feedback, and Instructional Intervention (PI: **Ouyang Fan**, Zhejiang University). *Main participant*
 - 2019.01 – 2022.12. National Natural Science Foundation of China (NSFC), Young Scientists Fund (Grant No. 61907038). Online Collaborative Knowledge Innovation among University Students: Development and Application of Real-Time Learning Analytics Tools (PI: **Ouyang Fan**, Zhejiang University). *Main participant*

- 2022.01 – 2023.12.Zhejiang Provincial Social Sciences Planning Project (Grant No. 2022SCG 256). Development and Empirical Investigation of an Intelligent Collaborative Teaching Model in Higher Education (PI: [Ouyang Fan](#), Zhejiang University). *Main participant*
- 2022.01 – 2023.12.National Social Science Fund of China, Major Project in Education (Grant No. VFA220003). Research on the Role of Education in Promoting Common Prosperity in the New Development Stage (PI: [Gu Jianmin](#), Zhejiang University). *Main Participant*
- 2022.01 – 2022.12. Zhejiang University–University College London Strategic Partnership Project. Improving Education Quality and Equity under the COVID-19 Pandemic: The Design and Implementation of Multimodal Data-Driven Artificial Intelligence in Collaborative Learning (PI: [Mutlu Cukurova](#), University College London). *Main Participant*
- 2024.01 – 2026.12. Ministry of Education Strategic Research Fund (MOE/SRF, Singapore). A multimodal learning analytical approach to highlight group engagement and collaboration for learning (PI: [Teo Wei-Peng](#), Nanyang Technological University). *Main participant*

Honors and Awards

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| 2025 | Doctoral Student Rising Star Training Programme, Zhejiang University (Totaling 87,000 RMB over three years). |
| 2025 | Junsheng Interdisciplinary Scholarship, Zhejiang University. |
| 2025 | Outstanding Graduate Student, Zhejiang University. |
| 2024 | National Construction High-Level University Postgraduate Program Scholarship, China Scholarship Council (Totaling approximately 140,000 RMB over 12 months). |
| 2024 | 5 th Zhang Dingzhang/Tong Miaoqin Scholarship, College of Education, Zhejiang University |
| 2024 | Outstanding Graduate Student, Zhejiang University. |
| 2023 | Ph.D. Excellent Training Scholarship (First Class), Zhejiang University. |
| 2023 | Excellent Academia Achievement Award, Third Prize, College of Education, Zhejiang University. |
| 2023 | 4 th Zhang Dingzhang/Tong Miaoqin Overseas Exchange Scholarship, College of Education, Zhejiang University. |
| 2022 | Certificate of Chu Kochen Honors Program, Zhejiang University. |
| 2022 | Outstanding Graduate of Zhejiang Province, Zhejiang Provincial Department of Education. |
| 2021 | 7 th Outstanding Achievement Award in Student Humanities and Social Sciences Research, Second Prize, Zhejiang University. |
| 2021 | Academic Excellence Model, Innovation and Entrepreneurship Excellence Model, Social Work Excellence Model, Zhejiang University. |
| 2020 | [Team leader] Shang De Public Service Scholarship, Zhejiang University. |
| 2019 | National Scholarship, Ministry of Education of the People's Republic of China. |

Service

Courses Taught

Data Science in Education (Undergraduate Course, Blended Course)

Teaching Assistant. Preparing course syllabus and teaching calendar; Drafting lecture slides; Organizing online and face-to-face group discussions; Supervising assignments; Student consultations.

Teaching Practicum in Education (Undergraduate Course)

Teaching Assistant. Practicum coordination; Placement coordination; Student supervision; Assignment grading.

Learning Analytics and Educational Data Mining (Graduate Course)

Teaching Assistant. Preparing lecture slides; Assignment design, feedback, and grading; Final project coordination.

Paper Reviewing

Computers & Education: Artificial Intelligence; Thinking Skills and Creativity; Computers & Education: X Reality

Invited Talks

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| 2022.5 | Invited talks on conducting educational research with learning analytics – Research experiences and practical insights for undergraduate students. Chu Kochen Honors College, Zhejiang University. |
| 2022.5 | Research presentation on examining the effects of a learning analytics tool on teaching and learning in small-class courses. University College London - Zhejiang University Strategic Project Symposium |
| 2022.6 | Invited talks on implementing data-driven learning and teaching analytics in higher education – Symposium on interim findings from teaching and learning research projects. Office of Undergraduate Education, Zhejiang University. |